

SHUBHAM KUMAR SHAW

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GitHub | LinkedIn | HDL-Bits

Durgapur, West Bengal – 713211, India

EDUCATION

Heritage Institute of Technology

Bachelors of Technology, Electronics and Communication Engineering (ECE), CGPA: **9.56 / 10.0** Kolkata, West Bengal

Durgapur Public School

12th Central Board of Secondary Education, Score: **91.2%** Durgapur, West Bengal

Durgapur Public School

10th Central Board of Secondary Education, Score: **95.4%** Durgapur, West Bengal

SKILLS

VLSI | Frontend Design

• Verilog HDL • VHDL • RTL Design & FSM Modelling • Functional Simulation • Waveform Analysis

VLSI | Physical Design

• RTL-to-GDSII Flow • Sky 130 Open PDK

EDA Tools

• Xilinx Vivado • Cadence Virtuoso • OpenSTA • OpenLane2 • Yosys • OpenROAD

Scripting

• TCL (Basics)

Professional Skills

• Communication Skills • Team Collaboration • Microsoft Office (Excel, Word, PowerPoint)

PROJECTS

• 5-Stage 32-bit Pipelined RISC-V Processor | Verilog HDL | OpenLane2 | Sky130 PDK

- Designed and implemented a 32-bit 5 staged pipelined RISC-V CPU in Verilog HDL with IF, ID, EX, MEM and WB stages using pipelined registers.
- Executed complete RTL-to-GDSII ASIC Implementation flow using OpenLane2 and Sky130 Open PDK including synthesis, floorplanning, placement, clock tree synthesis, routing, timing analysis, DRC/LVS verification and final GDSII generation.

• UART Communication Controller | Verilog HDL | Vivado

- Designed and verified UART Transmitter and Receiver with baud-rate generator using Verilog for serial communication.
- Implemented loopback architecture and validated successful data transmission/reception through behavioral simulation and waveform analysis in Vivado.

• 4-bit Kogge Stone Adder | Full-Custom ASIC Design | Cadence Virtuoso – Major Final Year Project

- Designed and implemented a 4-bit transistor-level Kogge Stone Adder in Cadence Virtuoso using hierarchical full-custom design methodology.
- Performed schematic capture, simulation, layout, DRC/LVS verification, and GDSII generation.

EXPERIENCE

• Industrial Trainee | Steel Authority of India Limited (DSP) - On-Site

13th July 2026 – 24th July 2026

- Gained hands-on exposure to industrial instrumentation, process control, PLC/SCADA systems and electronic control systems during industrial training at SAIL.

• VLSI Design Intern | Euphoria GenX – On-Site

3rd June 2026 – 13th July 2026

- Designed and implemented VHDL-based digital circuits using Behavioral, Dataflow, and Structural modeling, including a Digital Clock, Traffic Light Controller, Ripple Carry Adder, Up/Down Counter, Adders, and Subtractors in Xilinx Vivado.
- Performed RTL design, testbench development, functional simulation, waveform verification, RTL synthesis, FSM design, and debugging, following the complete FPGA design flow from specification to implementation.

• VLSI Design Intern | VAIDSYS Technologies – Remote

6th May 2026 – 5th June 2026

- Designed and implemented RTL Projects including 32-bit ALU, Pipelined RISC-Processor, SRAM Memory and Static Timing Analysis (STA)
- Performed functional simulation, waveform analysis, synthesis, timing verification and optimization of digital circuits.

POSITIONS OF RESPONSIBILITY

- Event Coordinator: IEEE MTT- S Smart System Design Challenge 2025 • Team Leader: IEEE MTT- S SBC Graphic Design Team (2023 – 2025) • Design Team Lead: HULT 2025

CERTIFICATIONS

- VLSI Microchip Design and Methodology – IEEE EDS SBC HITK • TCS-iON Career Edge – Young Professional

